

A5 · Answer Key and Demo Script

CS425 · Errors on Purpose · instructor led

INSTRUCTOR COPY

Every transcript below is real output captured on `onyx.boisestate.edu` on October 4, 2026. Hops and times will drift; the shapes will not. About 40 minutes: 8, 8, 8, 12, then 4 for the exit question.

Say this at the start

"In A4, tracepath counted routers by getting them to throw our packets away and complain about it. Those complaints are ICMP. Today we send ICMP's most famous message and see how far it gets. Everyone `ssh onyx`. Predict before we run anything." Then hand out the sheet.

ROUND 1 · IS ANYBODY OUT THERE?

```
ping -c 3 8.8.8.8 then ping -c 3 132.178.227.1 then curl ... https://www.google.com
```

```
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2053ms

64 bytes from 132.178.227.1: icmp_seq=1 ttl=254 time=0.603 ms
...
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
200
```

SAY "Is Google down? Is Onyx off the Internet? Neither. Something between here and there does not like ping."

Q1. Nothing is broken. The gateway answers echo requests, and TCP to port 443 at Google works (200). ICMP echo requests leaving campus are filtered, while TCP 443 is allowed. Ping tests one protocol on one path; it is not a test of "the network". Filtering by protocol and port is exactly what a middlebox does (4.5).

ROUND 2 · TRACEROUTE BY HAND

```
ping -c 1 -t 1 8.8.8.8 then -t 2, -t 3
```

```
From 132.178.227.1 icmp_seq=1 Time to live exceeded
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms

1 packets transmitted, 0 received, 100% packet loss, time 0ms (-t 2)
1 packets transmitted, 0 received, 100% packet loss, time 0ms (-t 3)
```

TTL	WHAT CAME BACK
-t 1	ICMP time exceeded (type 11, code 0) from the gateway, 132.178.227.1
-t 2	Nothing
-t 3	Nothing

Q2. The gateway: From 132.178.227.1 is the source address of the ICMP message, and the router that drops the datagram is the one that writes it. Silence does **not** mean no router: round 4 shows a router at hop 3 that answers UDP probes. Either hop 2 does not send the message (or rate limits it), or the echo request never reached hop 3. Round 4 settles which.

ROUND 3 · HOW DO YOU KNOW YOU ARRIVED?

```
python3: UDP datagram to 198.60.22.13 port 33434, then recv with a 3 second timeout
```

```
Traceback (most recent call last):
  File "<stdin>", line 6, in <module>
ConnectionRefusedError: [Errno 111] Connection refused
```

SAY "It came back in a few milliseconds, not 3 seconds. Somebody answered."

Q3. Nothing was listening on port 33434, so XMission's server answered the UDP datagram with ICMP **port unreachable** (type 3, code 3). Onyx's kernel matched the ICMP message to the socket (the ICMP payload carries the original IP header and the first 8 bytes of the UDP header, so the ports are in it) and reported it on the next call as **ECONNREFUSED**. That only works because the socket is `connect()`ed. `127.0.0.1` gives the same error, from Onyx itself. This is how `tracepath` printed reached in A4.

ROUND 4 · SAME PATH, THREE WAYS

mtr -n -r -c 5 mirrors.xmission.com (ICMP, the default)								
1. -- 132.178.227.1	0.0%	5	0.6	0.6	0.6	0.7	0.1	
2. -- 10.1.2.53	60.0%	5	0.5	0.5	0.5	0.5	0.0	
3. -- ???	100.0	5	0.0	0.0	0.0	0.0	0.0	

mtr -n -u -r -c 5 mirrors.xmission.com (UDP; TCP with -T -P 443 looks the same)								
1. -- 132.178.227.1	0.0%	5	0.7	0.6	0.5	0.7	0.1	
2. -- ???	100.0	5	0.0	0.0	0.0	0.0	0.0	
3. -- 132.178.1.147	0.0%	5	0.8	0.8	0.8	0.9	0.1	
4. -- 132.178.1.98	0.0%	5	1.6	1.6	1.4	1.7	0.1	
5. -- 209.249.232.13	0.0%	5	1.6	1.5	1.2	1.7	0.2	
6. -- ??? ... 9. -- ???	100.0							
10. -- 4.69.220.109	0.0%	5	27.9	27.8	27.3	28.1	0.3	
11. -- 4.35.168.186	0.0%	5	52.2	52.2	52.1	52.3	0.1	
12. -- 166.70.1.2	0.0%	5	52.3	52.2	52.1	52.4	0.1	
13. -- 198.60.22.13	0.0%	5	52.2	52.1	52.1	52.2	0.0	

SAY "Same path, same moment, three different answers. The probe you pick changes what you see."

PROBE	LAST HOP THAT ANSWERED	HOP 2
ICMP echo	hop 2 (the trace stops at 3)	10.1.2.53, 2 of 5 probes
UDP	hop 13, the server	???
TCP 443	hop 13, the server	??? (answered once in rehearsal)

Q4. Hop 2, or something right behind it. Hop 3 answers UDP and TCP probes every time, and answers a direct ping, so it would almost certainly have answered a TTL-expired echo request too if one had reached it. None did, so echo requests headed off campus are dropped at hop 2. Hop 2's address is `10.1.2.53`, an RFC 1918 private address: a box inside campus that is not meant to be reached from outside. Probably the campus firewall, but that is not confirmed, so say "probably".

Q5. Forwarding and answering are different jobs. Forwarding is the data plane, done in hardware for every packet. Making an ICMP time exceeded message is work for the router's own CPU, the control plane, and routers rate limit it, deprioritize it, or turn it off. A hop that loses 100% of the probes is a router that does not talk back, not one that drops your traffic. Loss only matters if it carries through to the last hop.

IF IT HAPPENS LIVE In one rehearsal the gateway took 1,032 ms to answer one TCP probe while hop 3 behind it answered in 0.7 ms. A router behind it cannot be faster at forwarding; the gateway was just slow to write the ICMP message. Same lesson as Q5.

EXIT QUESTION

Any two of: the gateway answers ping, so ICMP works locally; `curl` to Google returned `200`; UDP and TCP traces reached Salt Lake City; XMission's server answered round 3 with port unreachable; the ICMP trace dies at a private address inside campus, so the drop is local policy, not a dead Internet.

GRADING

Pass/fail, 20 points. Pass when every box has something in it. Wrong predictions are fine; that is what they are for.